

DAY 4

Metagenome- Assembled Genomes

Binning • MetaBAT2 • DAS_Tool • CheckM2 • MIMAG standards • GTDB-Tk taxonomy

What are Metagenome-Assembled Genomes (MAGs)?

"Near-complete genomes of individual microbial species, reconstructed by grouping (binning) assembled contigs from a metagenomic sample."



No culturing needed

Recover genomes of organisms that have never been grown in the lab. Access the "dark matter" of microbiology.



Go beyond taxonomy

Full gene content, metabolic pathways, virulence factors, AMR genes — not just "who is there" but "what can they do?"



New species discovery

Many MAGs represent novel genera and families with no cultured representatives. Rapidly expanding the tree of life.



Population genomics

Compare MAGs across samples to track strain-level diversity, horizontal gene transfer, and evolutionary dynamics.

MAG Reconstruction Pipeline

01 Assemble

MEGAHIT / SPAdes

De novo assemble reads into contigs (from Day 3).
Filter contigs < 1000 bp.

02 Map Reads

Bowtie2 + Samtools

Map original reads back to assembly. Coverage profiles used by binners.

03 Bin Contigs

MetaBAT2 / MaxBin2

Group contigs into bins (one bin $\approx$ one genome) using coverage + tetranucleotide composition.

04 Bin Refinement

DAS_Tool

Combine bins from multiple binners, select non-redundant, highest-quality set.

05 Quality Check

CheckM2

Estimate completeness and contamination of each MAG using marker gene sets.

06 Taxonomy

GTDB-Tk

Place MAGs into standardised GTDB phylogeny. Assign taxonomy to genus/species level.

SECTION 1

Binning

MetaBAT2 • Coverage • Tetranucleotide Composition

Binning: How MetaBAT2 Works

Binning groups assembled contigs into "bins" where each bin represents one genome.



Signal 1: Coverage Depth

Contigs from the same organism should have similar coverage across samples.

If organism A is 10× abundant in sample 1, all its contigs will also be ~10× in sample 1.

Coverage calculated by mapping reads back to assembly (Bowtie2 → Samtools).



Signal 2: Tetranucleotide Frequency

Each organism has characteristic nucleotide composition patterns — like a DNA "fingerprint".

256 possible 4-mer frequencies. Contigs from the same organism cluster together in 4-mer space.

PCA on 4-mer profiles reveals organism-specific clusters.

MetaBAT2 Workflow

1. Generate coverage from BAM file:
`jgi_summarize_bam_contig_depths --outputDepth depth.txt *.bam`
2. Run binning:
`metabat2 -i assembly.fasta -a depth.txt -o bins/bin --minContig 1500`

Bin Refinement: DAS_Tool

Different binners produce different (sometimes complementary) sets of bins. DAS_Tool selects the optimal, non-redundant set.

1 Run multiple binners

MetaBAT2, MaxBin2, and CONCOCT each produce their own set of bins from the same assembly. Each uses slightly different algorithms.

2 Score each bin

DAS_Tool uses single-copy marker genes to score each bin for completeness and contamination. Builds a scoring function.

3 Select best set

For overlapping bins (same contigs in multiple bins), DAS_Tool selects the highest-scoring one. Produces a non-redundant, optimal set.

4 Refined bin output

Typically 20–50% more high-quality MAGs recovered vs any single binner alone. Greatly reduces contamination.

```
DAS_Tool -i metabat2.scaffolds2bin,maxbin2.scaffolds2bin \  
-c assembly.fasta -o DAS_output --search_engine diamond --write_bins 1 -t 8
```

SECTION 2

MAG Quality Assessment

CheckM2 • Completeness • Contamination • MIMAG Standards

CheckM2 & MIMAG Quality Standards

CheckM2

Completeness (%)

Fraction of expected single-copy marker genes present. 90% = nearly complete genome.

Contamination (%)

Fraction of marker genes present in unexpected duplicates. <5% = low contamination.

Machine learning model

CheckM2 uses ML trained on GTDB genomes — faster and more accurate than CheckM1.

Strain heterogeneity

Contamination from same species vs different species. High SH = likely mixed strains.

```
checkm2 predict --input bins/ --output checkm2_results/ -x fa --threads 8
```

MIMAG Standards

Minimum Information about a Metagenome-Assembled Genome

(Bowers et al., 2017, Nature Biotechnology)

Tier	Complete	Contam.	Other
High Quality	≥ 90%	< 5%	tRNA + rRNA genes present
Medium Quality	≥ 50%	< 10%	No rRNA requirement
Low Quality	< 50%	< 10%	Preliminary use with caution

Note: Contamination % = (duplicated markers / total expected markers) × 100.

Aim for ≥ high quality MAGs for downstream functional analysis.

SECTION 3

MAG Taxonomy

GTDB-Tk • GTDB vs NCBI • Phylogenetic Placement

GTDB-Tk: Standardised Genome Taxonomy

How GTDB-Tk Works

Marker gene identification

Identify 120 bacterial / 53 archaeal conserved marker genes in each MAG using hmmsearch.

Tree placement

Place MAGs into the GTDB reference tree using maximum likelihood (pplacer/FastANI).

Taxonomy assignment

Assign taxonomy based on phylogenetic position relative to reference genomes.

ANI classification

Average Nucleotide Identity (ANI) $\geq 95\%$ = same species. Used to define species-level groups.

Novelty detection

MAGs outside reference clades flagged as potentially novel genera/families/phyla.

```
gtdbtk classify_wf --genome_dir bins/ --out_dir gtdbtk_output/ \
--cpus 8 --skip_ani_screen
```

GTDB vs NCBI Taxonomy

Feature	GTDB	NCBI
Basis	Phylogenomics (genome-based)	Historical / phenotypic
Stability	Standardised, reproducible	Changes with nomenclature
Novel organisms	Placed systematically	Often no taxonomy
Archaea	Major reclassifications	Traditional classification
Database	GTDB r220 (Apr 2024)	RefSeq / GenBank
Use case	MAG taxonomy, tree placement	Standard reference

GTDB reclassifies ~30% of bacteria and many archaea relative to NCBI. Always specify which taxonomy you used in publications!

Downstream: MAG Statistics & Visualisation

MAG Quality Summary Table

Compile CheckM2 completeness, contamination, contig count, N50, genome size for all MAGs. Filter by MIMAG tier. Summary statistics by quality tier.

R: dplyr + knitr / Excel

Taxonomy Distribution

Stacked bar plot of MAG taxonomy at phylum / family level. Compare across samples. Identify novel (uncultured) taxa with "UBA" or "GCA_" GTDB prefixes.

R: ggplot2, GTDB-Tk output

Coverage & Abundance

Map reads to MAG sequences. Calculate mean coverage and relative abundance per sample. Cluster MAGs by abundance profiles across samples.

CoverM + R heatmap

Functional Annotation Summary

After Prokka/eggNOG annotation, compare functional profiles across MAGs. KEGG pathway coverage, COG category distribution.

R: ggplot2, eggNOG output (Day 5)

Tip: Use the GTDB-Tk summary file (gtdbtk.bac120.summary.tsv) as the starting point for all downstream taxonomy work.

Today's Hands-On Session

1 Map Reads & Get Coverage

Map trimmed reads back to assembly with Bowtie2. Convert to BAM with Samtools. Generate contig depth file for MetaBAT2.

2 Bin with MetaBAT2

Run MetaBAT2 on the assembly. Run MaxBin2 for comparison. Count bins produced by each binner and compare.

3 Refine with DAS_Tool

Combine MetaBAT2 and MaxBin2 bins using DAS_Tool. Compare bin count and quality before and after refinement.

4 CheckM2 + GTDB-Tk

Run CheckM2 on refined bins — classify by MIMAG quality tier. Run GTDB-Tk to assign taxonomy. Build summary table in R.

Day 4 Summary

MAGs

Near-complete genomes reconstructed from metagenomics — enabling culture-independent genomics.

Binning

Groups contigs by coverage depth + tetranucleotide composition. MetaBAT2 is fast and widely used.

DAS_Tool

Combines multiple binners and selects the optimal non-redundant set. Improves bin quality significantly.

CheckM2

Estimates completeness and contamination using marker genes. MIMAG: HQ = >90% / <5%.

GTDB-Tk

Places MAGs in GTDB phylogenetic framework using 120 marker genes and FastANI. Novel taxa detected.

Outputs

Summary table of all MAGs with taxonomy, quality metrics, and coverage — basis for Day 5 functional analysis.

Day 5 tomorrow → Functional Analysis + Mini-Project (*HUMAN3*, *Prokka*, *eggNOG*, *AMR genes*)

KEY REFERENCES

Kang et al. (2019)

MetaBAT 2: adaptive binning algorithm for metagenome assembly

PeerJ 7: e7359

Sieber et al. (2018)

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CheckM2: rapid, scalable assessment of genome quality using machine learning

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GTDB: ongoing census of bacterial and archaeal diversity

Nucleic Acids Research 50: D785–D794

Wu et al. (2016)

MaxBin 2.0: automated binning algorithm to recover multiple microbial genomes

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